Econ 607 Notes

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Simple Endowment Economy

Primitives

- Last for $T+1$ periods, T is odd
- One good, perishable (i.e., no storage)
- Two types of consumers: A, B, of which there are N^A and N^B
- Endowments: $A : \{1, 0, 1, ..., 0\}, B : \{0, 1, 0, ..., 1\}$
- Utility: $\sum_{t=0}^{\infty} \beta^t u(c_t^j)$ (usually log utility)

Def "real savings"

$$b_t^j = \frac{s_t^j}{p_t}$$

Def "real rate of return (aka Fisher Equation)"

$$1 + r_t = (1 + i_t)\frac{p_t}{p_{t+1}}$$

Can also write

$$1 + r_t = \frac{1 + i_t}{1 + \pi_t} \Rightarrow \underbrace{r_t \approx i_t - \pi_{t+1}}_{\substack{\text{by logs:} \\ \text{approx in discrete} \\ \text{exact in cont time}}}$$

Budget constraints:

- period 0: $\underbrace{p_0 y_0^j}_{\text{sell all}} \geq \underbrace{p_0 c_0^j}_{\text{buy some back}} + \underbrace{s_0^j}_{\text{save rest}}$
- period t: $p_t y_t^j + s_{t-1}\underbrace{(1 + i_{t-1})}_{\substack{\text{1+net =} \\ \text{gross nom}}} \geq p_t c_t^j + s_t^j$
- period T: $p_T y_T^j + s_{T-1}(1 + i_{T-1}) \geq p_T c_T^j + \underbrace{s_T^j}_{\text{need no Ponzi}}$
- No Ponzi: $s_T^j \geq 0$
- rewrite period t to convert from nominal: $y_t^j + \frac{s_{t-1}}{p_t}(1 + i_{t-1}) \geq c_t^j + \frac{s_t^j}{p_t}$
- rewrite period t in terms of real savings: $y_t^j + \underbrace{\frac{s_{t-1}}{p_{t-1}}}_{b_{t-1}} \frac{p_{t-1}}{p_t}(1 + i_{t-1}) \geq c_t^j + \underbrace{\frac{s_t^j}{p_t}}_{b_t}$
- Final version of budget constraint:

$$y_t^j + b_{t-1}^j(1 + r_{t-1}) \geq c_t^j + b_t^j$$

or, for period 0:

$$y_0^j \geq c_0^j + b_0^j$$

Def "Competitive equilibrium"

A *competitive equilibrium* consists of

- prices $\{1 + r_t\}_{t=0}^{T-1}$ and
- sequences of allocations $\{c_t^A, c_t^B, b_t^A, b_t^B\}_{t=0}^{T}$ such that

1. Taking prices as given, the sequences of allocations maximize utility for $j = A, B$, and
2. All markets clear:

- goods market: $N^A c_t^A + N^B c_t^B = N^A y_t^A + N^B y_t^B$
- money/bond market: $N^A b_t^A + N^B b_t^B = 0$

Def "standard Euler Equation"

$$c_{t+1} = \beta(1 + r_t)c_t$$

Remember: this EE corresponds to a single person

Dynamic Programming

Consumption

Def "PIH"

As soon as income is promised, lifetime consumption should increase proportionally over each period.

Variations on PIH

1. Precautionary Savings
2. Borrowing Constraints
3. "Irrational" Intertemporal Choice

1. Precautionary Savings

Assume $1 + r = \frac{1}{\beta}$. Start with Euler Equation:

$$u'(c_t) = E_t[u'(c_{t+1})]$$

2nd order Taylor Approximate to get:

$$u'(c_t) \approx E_t[u'(c_t) + u''(c_t)(c_{t+1} - c_t) + \frac{u'''(c_t)}{2!}(c_{t+1} - c_t)^2]$$

Note: you can pull anything dated t or earlier out of expectation. Cancel, rearrange, divide by c_t to get

$$E[\frac{c_{t+1} - c_t}{c_t}] \approx \left(-\frac{1}{2}\right)\left(\frac{u'''(c_t)}{u''(c_t)}c_t\right)\left(E_t\left[\left(\frac{c_{t+1} - c_t}{c_t}\right)^2\right]\right)$$

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2. Borrowing Constraints

Def "primitives of Stochastic Exchange Economy"

- endowments
- no storage
- mass 1 of consumers
- $y_t(i) \sim \varphi[y^{min}, y^{max}]$ (this is a pdf or pmf that characterizes endowments. Importantly: endowments not correlated across individuals)

Claim:

$$Y_t = \int_0^1 y_t(i)di = \bar{y}$$

Def "stationay stochastic equilibrium"

A *stationay stochastic equilibrium* consists of

- a (single) interest rate $(1 + r)$,
- a policy function $h(y, A)$, and
- a distribution $g(A)$,

such that

1. $h(\cdot)$ is optimal given $1 + r$,
2. All markets clear
3. $g(A)$ is implied by $h(\cdot)$.

   individuals may move around, but the distribution itself is stationary

3. "Irrational" Intertemporal Choice

Commitment devices...

Laibson (1997)...

quasi-hyperbolic discounting...

Compare $t + k$ vs $t + k + 1$:

At time t:

$$MRS = \frac{\delta\beta^{t+k}u(c_{t+k})}{\delta\beta^{t+k+1}u(c_{t+k+1})} = \frac{u'(c_{t+k})}{\beta u(c_{t+k+1})}$$

At time $t + k$:

$$MRS = \frac{\beta^{t+k}u(c_{t+k})}{\delta\beta^{t+k+1}u(c_{t+k+1})} = \frac{u'(c_{t+k})}{\delta\beta u(c_{t+k+1})}$$

Leahy (2004) Valuation of past consumption...

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